COMPREHENSIVE RESEARCH DOSSIER

Project Codename: NIRMAAN / ChainTrace AI
SIH Problem Statement: SIH26183 — *Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics*
Issuing Authority: Ministry of Home Affairs (MHA) / Indian Cybercrime Coordination Centre (I4C)
Document Version: 1.0.0

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1. Problem Statement Context & The Indian Cybercrime Landscape

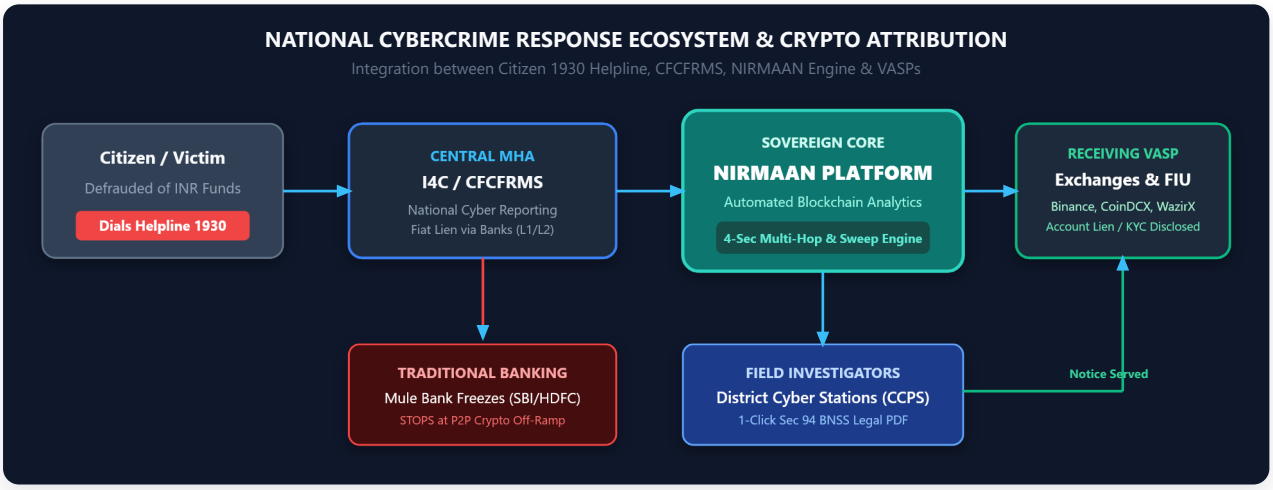


Figure 3.1: National Cybercrime Response Ecosystem & Automated Crypto Attribution Architecture

1.1 The Operational Gap in the 1930 Helpline & CFCFRMS

The Indian Cybercrime Coordination Centre (I4C), under the Ministry of Home Affairs (MHA), operates the **Citizen Financial Cyber Fraud Reporting and Management System (CFCFRMS)** and the national toll-free helpline **1930**.

* **The "Golden Hour":** When a victim reports a cyber fraud within 2 hours, the 1930 system dispatches automated API freeze requests down traditional banking channels (Layer 1, Layer 2, Layer 3 mule bank accounts).
* **The Crypto Black Hole:** Cybercrime syndicates (specifically Southeast Asian syndicates in Cambodia, Myanmar, and Laos operating "pig butchering", task fraud, and digital arrest scams) convert stolen INR into cryptocurrency via P2P desks on exchanges within **15 to 30 minutes**.
* Traditional banking liens hit an empty account; the money has already exited into **Tether (USDT) on TRON (TRC-20) or Binance Smart Chain (BEP-20)**.

1.2 Official Fraud Statistics (2024–2025 Benchmarks)

- **National Loss Volume:** Indian citizens lost **₹22,845 Crore** to cyber financial fraud in calendar year 2024.
- **Complaint Volume:** Total complaints registered on the National Cybercrime Reporting Portal (NCRP) crossed **65.89 Lakh cumulative complaints**, with cumulative reported citizen losses exceeding **₹55,050 Crore**.
- **Crypto's Laundering Share:** I4C empirical studies indicate that **over 35% of all high-value cyber fraud proceeds** are laundered via USDT on TRC-20 and EVM chains, representing an estimated **₹7,000 to ₹9,000 Crore of annual illicit crypto outflows from India**.

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2. In-Depth Competitive Intelligence

2.1 Competitor Benchmarking Matrix

COMPETITOR	VALUATION	EST. ANNUAL REVENUE (ARR)	LICENSE COST / SEAT	TARGET SEGMENT	KEY SHORTCOMINGS IN INDIAN CONTEXT
Chainalysis	$8.6B	~$175M – $200M+	**$30,000 – $100,000+ / yr** (~₹45L–₹85L)	Federal LEAs (FBI, IRS-CI), Tier-1 Banks	Prohibitive cost; complex manual UI; zero native integration with I4C / BNSS legal notices.
TRM Labs	$2.0B	~$100M	**$35,000 – $80,000 / yr** (~₹30L–₹65L)	US Federal LEAs (ICE, Secret Service), Top VASPs	High-end pricing; indirect exposure scoring creates noise for frontline Indian officers.
Elliptic	$670M	~$35M – $65M	**$15,000 – $50,000 / yr** + API consumption	UK/EU Banks, Custodians, Regulators	Focused on Western TradFi compliance; weak dynamic sweep attribution on TRON.
Merkle Science	$100M–$250M	~$5M – $12M	**$10,000 – $30,000 / yr** base	Mid-market VASPs, regional cyber units	Slower graph visualization; lacks automated Section 94 notice dispatch workflows.
Bitquery	~$50M	~$3.2M	Developer API ($13/mo – $5k/mo)	Web3 developers, quants	Developer API tool; provides no out-of-the-box investigative GUI or court-ready dossiers.
MistTrack	Private	Undisclosed	Freemium bot + API credits	Incident response, DeFi security	Hong Kong/APAC focus; no Indian FIU-IND regulatory or statutory notice workflows.

2.2 Empirical Indian Police Procurement Evidence

- **Kerala Police Cyber Operations Tender (** `2024_KP_678291_1` **/** `E1(c)-6143/2024/Cyb(1)` **):**
- Procurement of **01 single-user license** of a cryptocurrency analyzing tool (Chainalysis).
- Sanctioned budget: **₹50.00 Lakhs ($60,000 USD)**; final award: **₹45.00 Lakhs**.

- **UP ATS & Bihar Police GeM Bids (`GEM/2025/B/6908820` , `GEM/2026/B/8076690`):**
- Single-station software tenders consistently exhaust ₹40L – ₹1.2 Crore, proving that foreign software is mathematically unaffordable for the 800+ subordinate district-level cyber police stations across India.

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3. Statutory Legal Framework (Active July 1, 2024)

To ensure evidence gathered by NIRMAAN is admissible in Indian courts, the platform natively implements the **new criminal law codes** enacted on July 1, 2024:

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4. Production-Ready Legal Notice Templates

4.1 Section 94 BNSS, 2023 Notice Template (To VASP Nodal Officer)

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FORM OF ORDER / NOTICE UNDER SECTION 94 OF THE BHARATIYA NAGARIK SURAKSHA SANHITA, 2023
(Corresponding to Section 91 of the Code of Criminal Procedure, 1973)

To:
The Nodal Officer / Compliance Department / Law Enforcement Response Team,
M/s [Name of Exchange, e.g., Binance / Bybit / CoinDCX / WazirX],
Registered VASP Legal Desk.

SUBJECT: REQUISITION FOR PRODUCTION OF RECORDS, TRANSACTION LEDGERS, AND URGENT FREEZING OF
         IN CONNECTION WITH INVESTIGATION OF FIR NO. [_______/2026] UNDER SECTIONS 316(2),
         AND SECTION 66D INFORMATION TECHNOLOGY ACT, 2000.

Whereas, an investigation into a cyber financial fraud complaint involving an amount of INR
(Equivalent to [________] USDT) is being conducted by the undersigned Investigating Officer

And whereas, automated blockchain forensic analysis of the fund trail originating from the
[0x_Victim_Address / T_Victim_Address] has established that illicit proceeds were deposited
address operated and maintained by your exchange:

1. Target Deposit Address: [Target_Deposit_Address]
2. Incoming Transaction Hash (TxID): [TxHash_Value]
3. Transaction Timestamp: [YYYY-MM-DD HH:MM:SS UTC]
4. Amount Deposited: [________] USDT
5. Parent Sweeping / Gas Dispenser Hash: [Sweep_TxHash]

Therefore, by virtue of powers conferred under Section 94 of the Bharatiya Nagarik Suraksha
you are hereby legally directed to produce the following records within 24 hours of receipt

1. Complete User Profile & KYC Records: Legal name, DOB, residential address, government ID
   registered phone number, registered email address, and IP/facial verification records.
2. Complete Transaction Ledger: Comprehensive deposit, order-book execution, and withdrawal
   associated with the account UID linked to the above deposit address.
3. Device & Access Telemetry: IP logs with port numbers, timestamps, login history, IMEI/MA
4. Linked Banking / P2P Settlement Records: Bank accounts (Account number, IFSC, Bank name)
   registered for P2P counterparty fiat off-ramping.
5. Immediate Preservation & Lien (Section 106/107 BNSS): Place an immediate freeze on all a
   held under the target UID and restrict all outgoing crypto withdrawals and fiat settleme

Failure to comply renders you liable for penal action under Section 223 of the Bharatiya Ny

Given under my hand and seal of the Police Station, this [Day] day of [Month], 2026.

Investigating Officer (IO): _____
Rank / Designation: _____
Police Station: _____
State / District: _____
Official Email (gov.in/nic.in): _____
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4.2 Section 63 BSA, 2023 Electronic Evidence Certificate

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CERTIFICATE UNDER SECTION 63(4)(c) OF THE BHARATIYA SAKSHYA ADHINIYAM (BSA), 2023
(Corresponding to Section 65B of the Indian Evidence Act, 1872)
FOR ADMISSIBILITY OF ELECTRONIC FORENSIC BLOCKCHAIN RECORDS

I, [Name of Officer / Forensic Analyst], functioning at [Cyber Police Station / Organizatio
do hereby solemnly affirm and state on oath as under:

1. I am authorized and responsible for the technical extraction, on-chain traversal, and fo

PART A: IDENTIFICATION OF ELECTRONIC RECORD & SOURCE SYSTEMS
2. The electronic record produced herewith consists of a computer-generated forensic audit
   pages [1 to N], titled "BLOCKCHAIN TRANSACTION AUDIT REPORT – CASE FIR NO. [______/2026]
3. The electronic record was generated through automated blockchain inspection software ing
   header data and transaction logs via authenticated API interfaces connected to public de
   [TRON / Ethereum / BNB Chain].
4. Date & Time of Extraction: [YYYY-MM-DD HH:MM:SS IST]
5. Cryptographic Integrity Hash (SHA-256) of Raw Trace Ledger:
   [e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855]

PART B: CERTIFICATE REGARDING REGULAR AND ACCURATE OPERATION
6. In compliance with Section 63(2) of the Bharatiya Sakshya Adhiniyam, 2023, I certify tha
   (a) The computer system, database, and querying software used to compile this record ope
   (b) The electronic records were fetched in the ordinary course of legitimate forensic ar
       from immutable public blockchain distributed ledgers.
   (c) The record was protected against unauthorized access, data alteration, or malicious
   (d) The attached printout/PDF is an accurate reproduction of electronic ledger records s

IN WITNESS WHEREOF, I have signed this certificate on this [____] day of [________], 2026 a

Deponent / System Operator: _____________________________________
Designation: ____________________________________________________
Government ID / Badge No: _______________________________________
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5. Go-To-Market & Government Procurement (GeM Portal)

5.1 The GeM Portal Strategic Advantage

Under the Government of India's **Public Procurement (Preference to Make in India) Order (DPIIT)**:

* **Class-I Local Supplier:** Platforms developed in India with $> 50\%$ local value addition receive mandatory procurement preference in all state and central government tenders.
* **Overcoming Rule 166 GFR (General Financial Rules):** Procuring foreign software like Chainalysis requires complex single-source exemptions and foreign exchange permissions. An indigenous solution listed on the **GeM Portal (Government e-Marketplace)** can be directly procured by State Police Cyber Commands via L1 bidding or custom RFP categories under the **Modernisation of Police Forces (MPF) Scheme**.